

THE IDENTITIES OF A GROUP WITH NILPOTENT COMMUTATOR SUBGROUP ARE FINITELY BASED

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ABSTRACT. It is shown that the identities of a group with nilpotent commutator subgroup are finitely based. It follows that the identities of a connected matrix group over a field are finitely based.

Introduction

Soon after the finite basis problem for group identities was solved in the negative by Ol'shanskii [1], Adyan [2] and Vaughan-Lee [3], the following question was posed [3]. Does every group with nilpotent commutator subgroup have a finite basis for its identities? The importance of these groups stems mainly from Platonov's observation [4] that every matrix group over a field that satisfies a nontrivial identity is soluble-by-finite, and thus, by the Kolchin-Mal'tsev theorem, is a finite extension of a group with nilpotent commutator subgroup.

Cohen [5] proved that the identities of every metabelian group (group with abelian commutator subgroup) are finitely based, using the techniques of Higman [6]. Cohen's result has been strengthened successively in articles of Vaughan-Lee [7], McKay [8], and Bryant and Newman [9]; finally, it was proved in [9] that the identities of a group whose commutator subgroup is nilpotent of class 2 are finitely based, as well as those of some other groups. The methods of [9] are used in [10] to show that the identities of every extension of a nilpotent group by an abelian group of finite exponent are finitely based; in particular, this is true for supersoluble groups.

The main result of this paper is as follows.

THEOREM 1. *The identities of every group with nilpotent commutator subgroup are finitely based.*

Since every Zariski-connected matrix group over a field satisfying a nontrivial identity is soluble (see [4]), and every connected soluble linear group has nilpotent commutator subgroup, the theorem has the following consequence.

COROLLARY. *The identities of every connected linear group over a field are finitely based.*

The method developed in the proof of Theorem 1 can be used in certain cases to investigate the finite basis problem for representations of groups (for the definition of identities of a group representation and associated concepts, see [11]–[13]).

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THEOREM 2. *Every group representation over a Noetherian commutative associative ring K with 1 and satisfying the identity*

$$(x_1 x_2 - x_2 x_1) \cdots (x_{2n-1} x_{2n} - x_{2n} x_{2n-1}) \equiv 0, \quad (1)$$

for some n has a finite basis for its identities.

This theorem solves a problem posed by Plotkin [13]. There is an earlier proof [24] of Theorem 2 in the case where K is a field of characteristic 0 or a prime p such that $p \geq n$.

We note that, as in [15], we can define an injective homomorphism from the lattice of varieties of associative algebras over an infinite field into the lattice of varieties of group representations over the same field. Since the image under this homomorphism of the variety of associative algebras defined by identity (1) is a variety of group representations satisfying (1), we get the following corollary to Theorem 2.

COROLLARY. *Every associative algebra over an infinite field satisfying (1) has a finite basis for its identities.*

For fields of characteristic 0, this was proved by Latyshev [16] and Genov [17]; for $n = 2$ and 3 it follows from results of Popov [18] and Chiripov [19].

Set $[x_1, x_2] = x_1^{-1} x_2^{-1} x_1 x_2$, and $[x_1, \dots, x_k] = [[x_1, \dots, x_{k-1}], x_k]$ for $k \geq 3$; let $F = F(\mathfrak{N}_c \mathfrak{A})$ be the free group of countably infinite rank of the variety $\mathfrak{N}_c \mathfrak{A}$ of all groups whose commutator subgroups are nilpotent of class at most c . As is well known, to prove Theorem 1 it is enough to establish that ascending chains of verbal subgroups of $F(\mathfrak{N}_c \mathfrak{A})$ break off, for each c .

Let $x_1, x_2, \dots$ be the free generators for F , and Φ the set of all monotone maps of the set $\mathbf{N}$ of natural numbers to itself (that is, maps such that $\varphi(k) < \varphi(l)$ when $k < l$). We also use Φ to denote the family of endomorphisms φ of F such that $\varphi(x_i) = x_{\varphi(i)}$ for all $i \in \mathbf{N}$. Let Θ be the set of endomorphisms θ_{kl} , $k < l$, of F such that $\theta_{kl}(x_l) = x_k x_l$ and $\theta_{kl}(x_i) = x_i$ for $i \neq l$. Finally, let Ψ be the family of endomorphisms

$$\psi = \psi_{k; g^{(1)}, n^{(1)}, \dots, g^{(k)}, n^{(k)}} \quad (k \in \mathbf{N}, g^{(l)} \in F, n^{(l)} \in \mathbf{Z})$$

of F such that

$$\psi(x_i) = x_i [x_i, g^{(1)}]^{n^{(1)}} \cdots [x_i, g^{(k)}]^{n^{(k)}}$$

for all $i \in \mathbf{N}$.

In fact, rather than Theorem 1 we prove a stronger result:

THEOREM 1'. *For every natural number c , ascending chains of normal subgroups of $F = F(\mathfrak{N}_c \mathfrak{A})$ admitting all the endomorphisms from Φ , Θ , and Ψ break off.*

Let $\mathcal{F}$ be the (absolutely) free group of countably infinite rank. The endomorphism sets Φ , Θ , and Ψ , can be defined for $\mathcal{F}$ in the same way as for $F(\mathfrak{N}_c \mathfrak{A})$. The following result is stronger than Theorem 2.

THEOREM 2'. *Let K be any Noetherian commutative associative ring with 1. Let I be the verbal ideal (that is, ideal closed under the endomorphisms of $K[\mathcal{F}]$ induced by endomorphisms of $\mathcal{F}$ of the group and $K[\mathcal{F}]$ generated by the left-hand side of (1). Then, ascending chains of ideals of $K[\mathcal{F}]/I$ admitting all the endomorphisms of $K[\mathcal{F}]$ induced by endomorphisms in Φ , Θ , and Ψ break off.*

We note that Cohen's proof [5] of the finite basis theorem for metabelian groups proceeds in two steps: first, the finite basis problem is reduced to proving that a

certain operator module over the ring of polynomials in countably infinitely many unknowns is Noetherian; then, using a technique developed by Higman [6] concerning partially well-ordered sets, it is proved that this module is indeed Noetherian. In the later papers [7]–[10], progress was achieved at the expense of applying Higman's technique directly to commutators of a special form and refusing to consider modules over polynomial rings.

In this paper, we have a “return to Cohen”: the proof of Theorem 1' follows the same plan as in [5], even though the realization of both parts of it requires considerably greater effort. In particular, at the second stage of the proof (that of establishing that the module over the polynomial ring is Noetherian), although we do in fact use the technique of partial well-orderings (in proving Lemma 2.1), other considerations have a fundamental part to play. Theorem 2' reduces to the same sort of proof that a module over a polynomial ring is Noetherian as does Theorem 1'.

§1. Theorem 1' : reduction to a proof that a module over a polynomial ring is Noetherian

1. Let $H = \gamma_c(F')$ be the last nontrivial term of the lower central series of the commutator subgroup of the free group $F = F(\mathfrak{M}_c \mathfrak{A})$. To prove Theorem 1' by an inductive argument, it is enough to show that ascending chains of normal subgroups contained in H admitting the endomorphisms from Φ , Θ , and Ψ break off.

The following lemma is easily established.

LEMMA 1.1. *The subgroup H of F is generated qua abelian group by all elements of the form*

$$[[x_{i_1}, x_{i_2}]^{g_1}, \dots, [x_{i_{2c-1}}, x_{i_{2c}}]^{g_c}],$$

where the x_i , $i \in \mathbb{N}$, are free generators of F , $i_1, \dots, i_{2c} \in \mathbb{N}$, and $g_1, \dots, g_c \in F$.

For g in F , let $\bar{g}$ denote the coset $g \cdot F'$. We note that when $\bar{g}_1 = \bar{g}_2$, we have

$$[x_{i_1}, x_{i_2}]^{g_1} = [x_{i_1}, x_{i_2}]^{g_2} \pmod{F''}.$$

Since F' is nilpotent of class c , the following lemma is an easy consequence of this.

LEMMA 1.2. *Suppose that $i_1, \dots, i_{2c} \in \mathbb{N}$, and let g_{1j} and g_{2j} be elements of F such that $\bar{g}_{1j} = \bar{g}_{2j}$ for $j = 1, \dots, c$. Then*

$$[[x_{i_1}, x_{i_2}]^{g_{11}}, \dots, [x_{i_{2c-1}}, x_{i_{2c}}]^{g_{1c}}] = [[x_{i_1}, x_{i_2}]^{g_{21}}, \dots, [x_{i_{2c-1}}, x_{i_{2c}}]^{g_{2c}}].$$

This lemma means that elements of the form

$$[[x_{i_1}, x_{i_2}]^{\bar{g}_1}, \dots, [x_{i_{2c-1}}, x_{i_{2c}}]^{\bar{g}_c}], \quad (2)$$

are correctly defined in F , for any $i_1, \dots, i_{2c} \in \mathbb{N}$ and $\bar{g}_1, \dots, \bar{g}_c \in F/F'$. It follows from Lemma 1.1 that these elements generate H as an abelian group.

Using the Magnus embedding of a free metabelian group in a wreath product ([20], Chapter 3, §6), it is not hard to prove the following three lemmas.

LEMMA 1.3. *For arbitrary $\varphi \in \Phi$, $i_1, i_2 \in \mathbb{N}$, and $g = \bar{x}_1^m \bar{x}_2^m \dots \in F/F'$,*

$$\varphi([x_{i_1}, x_{i_2}]^{\bar{g}}) = [x_{\varphi(i_1)}, x_{\varphi(i_2)}]^{\varphi(\bar{g})} \pmod{F''},$$

where $\varphi(\bar{g}) = \bar{x}_{\varphi(1)}^{m_1} \bar{x}_{\varphi(2)}^{m_2} \dots$.

LEMMA 1.4. For arbitrary $\theta_{kl} \in \Theta$, $(k < l)$, $i_1, i_2 \in \mathbf{N}$, and $\bar{g} = \bar{x}_1^{m_1} \bar{x}_2^{m_2} \dots \in F/F'$, modulo F'' we have

$$\theta_{kl}([x_{i_1}, x_{i_2}]^{\bar{g}}) = h,$$

where

$$h = \begin{cases} [x_{i_1}, x_{i_2}]^{\theta_{kl}(\bar{g})} \pmod{F''}, & \text{if } i_1, i_2 \neq l; \\ [x_k, x_{i_2}]^{\bar{x}_l^{\theta_{kl}(\bar{g})}} [x_l, x_{i_2}]^{\theta_{kl}(\bar{g})} \pmod{F''}, & \text{if } i_1 = l, i_2 \neq l; \\ [x_{i_1}, x_l]^{\theta_{kl}(\bar{g})} [x_l, x_k]^{\bar{x}_l^{\theta_{kl}(\bar{g})}} \pmod{F''}, & \text{if } i_1 \neq l, i_2 = l. \end{cases}$$

LEMMA 1.5. For all $k, i_1, i_2 \in \mathbf{N}$, $g^{(1)}, \dots, g^{(k)} \in F$, $n^{(1)}, \dots, n^{(k)} \in \mathbf{Z}$, and $\bar{g} \in F/F'$, we have

$$\begin{aligned} \psi_{k; g^{(1)}, n^{(1)}, \dots, g^{(k)}, n^{(k)}}([x_{i_1}, x_{i_2}]^{\bar{g}}) \\ = ([x_{i_1}, x_{i_2}] [x_{i_1}, x_{i_2}, g^{(1)}]^{n^{(1)}} \dots [x_{i_1}, x_{i_2}, g^{(k)}]^{n^{(k)}})^{\bar{g}} \pmod{F''}. \end{aligned}$$

Using the fact that F' is nilpotent of class c , one easily deduces the following lemmas from Lemmas 1.3–1.5.

LEMMA 1.6. For arbitrary $\varphi \in \Phi$, $i_1, \dots, i_{2c} \in \mathbf{N}$, and $\bar{g}_l = \bar{x}_1^{m_{l1}} \bar{x}_2^{m_{l2}} \dots \in F/F'$ ($l = 1, \dots, c$), we have

$$\varphi([x_{i_1}, [x_{i_2}]^{\bar{g}_1}, \dots, [x_{i_{2c-1}}, x_{i_{2c}}]^{\bar{g}_c}]) = [[x_{\varphi(i_1)}, x_{\varphi(i_2)}]^{\varphi(\bar{g}_1)}, \dots, [x_{\varphi(i_{2c-1})}, x_{\varphi(i_{2c})}]^{\varphi(\bar{g}_c)}],$$

where $\varphi(\bar{g}_l) = \bar{x}_{\varphi(1)}^{m_{l1}} \bar{x}_{\varphi(2)}^{m_{l2}} \dots$ ($l = 1, \dots, c$).

LEMMA 1.7. For arbitrary $\theta_{kl} \in \Theta$ ($k \leq l$), $i_1, \dots, i_{2c} \in \mathbf{N}$, and $\bar{g}_j = \bar{x}_1^{m_{j1}} \bar{x}_2^{m_{j2}} \dots \in F/F'$ ($j = 1, \dots, c$) we have

$$\begin{aligned} \theta_{kl}([x_{i_1}, x_{i_2}]^{\bar{g}_1}, \dots, [x_{i_{2c-1}}, x_{i_{2c}}]^{\bar{g}_c}) \\ = [[x_{i_1}, x_{i_2}]^{\theta_{kl}(\bar{g}_1)}, \dots, [x_{i_{2c-1}}, x_{i_{2c}}]^{\theta_{kl}(\bar{g}_c)}] \\ \times \prod_{q \in Q} [[x_{i_{q_1}}, x_{i_{q_2}}]^{\bar{g}_{q_1}}, \dots, [x_{i_{q(2c-1)}}, x_{i_{q(2c)}}]^{\bar{g}_{q_c}}], \end{aligned}$$

where $\theta_{kl}(\bar{g}_j) = \bar{x}_1^{m_{j1}} \dots \bar{x}_k^{m_{jk} + m_{jl}} \dots \bar{x}_l^{m_{jl}} \dots$ ($j = 1, \dots, c$), and the index set Q can be empty. If Q is not empty, then $\bar{g}_{qj} \in \{\theta_{kl}(\bar{g}_j), \bar{x}_l^{\theta_{kl}(\bar{g}_j)}\}$, and $i_{qj} \leq i_j$ for all q and j , and for each q there exists j' such that $i_{qj'} < i_{j'}$.

LEMMA 1.8. For arbitrary $k, i_1, \dots, i_{2c} \in \mathbf{N}$, $g^{(1)}, \dots, g^{(k)} \in F$, $n^{(1)}, \dots, n^{(k)} \in \mathbf{Z}$, and $\bar{g}_1, \dots, \bar{g}_c \in F/F'$ we have

$$\psi_{k; g^{(1)}, n^{(1)}, \dots, g^{(k)}, n^{(k)}}([x_{i_1}, x_{i_2}]^{\bar{g}_1}, \dots, [x_{i_{2c-1}}, x_{i_{2c}}]^{\bar{g}_c}) = [h_1^{\bar{g}_1}, \dots, h_c^{\bar{g}_c}], \quad (3)$$

where

$$h_l = [x_{i_{2l-1}}, x_{i_{2l}}] [x_{i_{2l-1}}, x_{i_{2l}}, g^{(1)}]^{n^{(1)}} \dots [x_{i_{2l-1}}, x_{i_{2l}}, g^{(k)}]^{n^{(k)}} \quad (l = 1, \dots, c).$$

As is well known (see [20], for example), the subgroup F'/F'' of F/F'' can be considered as a $\mathbb{Z}[F/F']$ -module by setting

$$h(\sum_i m_i \bar{g}_i) = \prod_{i=1}^k (g_i^{-1} h g_i)^{m_i}$$

for all $h \in F'/F''$, $\bar{g}^{(1)}, \dots, \bar{g}^{(k)} \in F/F'$, and $m_1, \dots, m_k \in \mathbb{Z}$ (we are using multiplicative notation in F'/F'' for expressing the group operation, and exponentiation to denote the action of elements of $\mathbb{Z}[F/F']$ on F'/F''). In module notation, it is easy to see that

$$[x_i, x_j, g]^n = [x_i, x_j]^{-n(1-\bar{g})} \pmod{F''},$$

so that, by Lemma 1.5,

$$\psi_{k; g^{(1)}, n^{(1)}, \dots, g^{(k)}, n^{(k)}}([x_{i_1}, x_{i_2}]^{\bar{g}}) = [x_{i_1}, x_{i_2}]^{(1-n^{(1)}(1-g^{(1)})-\dots-n^{(k)}(1-g^{(k)}))\bar{g}} \pmod{F''}.$$

With this notation, we can put

$$h_p = [x_{i_{2l-1}}, x_{i_{2l}}]^{1-n^{(1)}(1-g^{(1)})-\dots-n^{(k)}(1-g^{(k)})}$$

in (3) (clearly, elements of the form $[h_1, \dots, h_c]$ with $h_i \in F'/F''$ for $i = 1, \dots, c$, can be defined properly in H).

2. Let A^+ be the subsemigroup of F/F' generated by the elements $\bar{x}_i$ ($i \in \mathbb{N}$), and let H^+ be the subgroup of H generated by all commutators of the form (2) in which $\bar{g}_i \in A^+$ ($i = 1, \dots, c$). It follows from Lemmas 1.6–1.8 that H^+ admits all the endomorphisms from Ψ and Θ , and also all endomorphisms $\psi = \psi_{k; g^{(1)}, n^{(1)}, \dots, g^{(k)}, n^{(k)}}$ in Ψ for which $\bar{g}^{(i)} \in A^+$ for $i = 1, \dots, k$ (we shall denote the set of all endomorphisms ψ of this form by Ψ^+). We note that, for each $h \in H$, there exists $g \in F$ such that $h^g \in H^+$; thus, for any normal subgroups N_1 and N_2 of F contained in H , we have $N_1 = N_2$ if and only if $N_1 \cap H^+ = N_2 \cap H^+$. We shall show that ascending chains of normal subgroups of F contained in H^+ and admitting all the endomorphisms from Φ , Θ , and Ψ^+ break off; by the above remark, this will then show that such chains in H break off.

Let $R_c = \mathbb{Z}[z_{ij} | 1 \leq i \leq c, j \in \mathbb{N}]$ be the ring of polynomials with 1 and L_c the free module over R_c freely generated by the elements $w_{i_1 i_2 \dots i_{2c}}$, where $i_l \in \mathbb{N}$ for $l = 1, 2, \dots, 2c$.

We define a $\mathbb{Z}$ -linear map δ of L_c to H^+ such that

$$\delta(w_{i_1 \dots i_{2c}} z_{11}^{m_{11}} z_{12}^{m_{12}} \dots z_{21}^{m_{21}} z_{22}^{m_{22}} \dots z_{c1}^{m_{c1}}) = [[x_{i_1}, x_{i_2}]^{\bar{g}_2}, [x_{i_3}, x_{i_4}]^{\bar{g}_2}, \dots, [x_{i_{2c-1}}, x_{i_{2c}}]^{\bar{g}_c}],$$

where $\bar{g}_i = \bar{x}_1^{m_{i1}} \bar{x}_2^{m_{i2}} \dots$ ($i = 1, \dots, c$).

Let $\psi = \psi_{k; g^{(1)}, n^{(1)}, \dots, g^{(k)}, n^{(k)}}$ be an endomorphism from Ψ^+ , $\bar{g}^{(i)} = \bar{x}_1^{n_{i1}} \bar{x}_2^{n_{i2}} \dots$ elements of F/F' for $i = 1, \dots, c$, and p_ψ the polynomial in $\mathbb{Z}[z_i | i \in \mathbb{N}]$ given by

$$p_\psi(z_1, z_2, \dots) = 1 - n^{(1)}(1 - z_1^{n_{11}} z_2^{n_{12}} \dots) - \dots - n^{(k)}(1 - z_1^{n_{k1}} z_2^{n_{k2}} \dots).$$

We note that $p_\psi(z_1, z_2, \dots)$ runs over the set of all polynomials in $\mathbb{Z}[z_i | i \in \mathbb{N}]$ such that $p(1, 1, \dots) = 1$ as ψ runs over Ψ^+ . Suppose that $q_\psi \in R_c$, and

$$q_\psi = p_\psi(z_{11}, z_{12}, \dots) \dots p_\psi(z_{c1}, z_{c2}, \dots).$$

We shall also use the symbol q_ψ for the endomorphism of L_c induced by multiplication by q_ψ . Using the distributivity of multiplication over addition in R_c and of commutation over multiplication in the factors of the lower central series of a nilpotent group, it is not difficult to prove the following lemma.

LEMMA 1.9. *The following diagram is commutative for every ψ in Ψ^+ :*

$$\begin{array}{ccc} L_c & \xrightarrow{q_\psi} & L_c \\ \delta \downarrow & & \delta \downarrow \\ H^+ & \xrightarrow{\psi} & H^+ \end{array}.$$

We recall that Φ is the set of all monotone maps from $\mathbf{N}$ to itself; we shall also denote by Φ the set of endomorphisms φ of the algebra R_c and of semilinear transformations of L_c such that

$$\begin{aligned} \varphi(z_{ij}) &= z_{i(\varphi(j))} \quad (i = 1, \dots, c; j \in \mathbf{N}), \\ \varphi(w_{i_1 \dots i_{2c}} r) &= w_{\varphi(i_1) \dots \varphi(i_{2c})} \varphi(r) \quad (i_1, \dots, i_{2c} \in \mathbf{N}, r \in R_c). \end{aligned}$$

We used Θ to denote a certain set of endomorphisms θ_{ij} ($i < j$) of F ; we shall use Θ to denote the set of all endomorphisms θ_{ij} ($i < j$) of the ring R_c such that

$$\theta_{ij}(z_{kl}) = z_{kl} \quad (l \neq j), \quad \theta_{ij}(z_{kj}) = z_{ki} z_{kj}.$$

We denote by Θ^* the set of all $\mathbf{Z}$ -linear maps θ_{ij}^* ($i < j$) of L_c to itself defined by the equations

$$\theta_{ij}^*(w_{i_1 \dots i_{2c}} r) = w_{i_1 \dots i_{2c}} \theta_{ij}(r) + \sum_{q \in Q} w_{i_{q1} \dots i_{q(2c)}} r^{(q)}$$

for all $i_1, \dots, i_{2c} \in \mathbf{N}$ and every monomial

$$r = z_{11}^{m_{11}} z_{12}^{m_{12}} \dots z_{c1}^{m_{c1}} z_{c2}^{m_{c2}} \dots$$

with coefficient the identity of R_c . Here, the index set Q and the numbers $i_{q1}, \dots, i_{q(2c)}$ and m_{kl} are just as in the conditions of Lemma 1.7, while $r^{(q)}$ is a monomial in R_c such that

$$\delta(w_{i_{q1} \dots i_{q(2c)}} r^{(q)}) = [[x_{i_{q1}}, x_{i_{q2}}]^{\bar{g}_{q1}}, \dots, [x_{i_{q(2c-1)}}, x_{i_{q(2c)}}]^{\bar{g}_{qc}}]$$

for all $q \in Q$ (see Lemma 1.7).

The next result is clear.

LEMMA 1.10. *For all φ in Φ and all i and j in $\mathbf{N}$ with $i < j$, the following diagram are commutative:*

$$\begin{array}{ccc} L_c & \xrightarrow{\varphi} & L_c \\ \delta \downarrow & & \delta \downarrow \\ H^+ & \xrightarrow{\varphi} & H^+ \end{array} \quad \begin{array}{ccc} L_c & \xrightarrow{\theta_{ij}^*} & L_c \\ \delta \downarrow & & \delta \downarrow \\ H^+ & \xrightarrow{\theta_{ij}} & H^+ \end{array}$$

Let Θ be the set of all semilinear transformations θ_{ij} of L_c such that

$$\theta_{ij}(w_{i_1 \dots i_{2c}} r) = w_{i_1 \dots i_{2c}} \theta_{ij}(r)$$

for all $i_1, \dots, i_{2c}$ in $\mathbf{N}$ and r in R_c . Let $\tilde{R}_c$ be the subalgebra of R_c generated by all polynomials q_ψ , $\psi \in \Psi^+$. As we mentioned above, to prove the theorem it suffices to show that ascending chains of subgroups of H^+ admitting endomorphisms from Φ , Θ , and Ψ^+ break off. Since the inverse image in L_c of a subgroup of this sort is an $\tilde{R}_c$ -submodule (Lemmas 1.9 and 1.10) admitting maps from Φ and Θ^* , for a proof of the theorem it is enough to establish that ascending chains of $\tilde{R}_c$ -modules admitting maps from Φ and Θ^* break off. It is easy to show that this last assertion is a consequence of the following proposition.

PROPOSITION 1.11. *Ascending chains of $\tilde{R}_c$ -submodules of L_c admitting maps from Φ and Θ break off.*

This is how the situation turns out: in order to proceed by induction on c , what we need is not Proposition 1.11 but a more general assertion, to be stated later. It will be convenient to view R_c as a ring of polynomials in the unknowns $t_{ij} = z_{ij} - 1$ ($i = 1, \dots, c$; $j \in \mathbf{N}$). It is clear that all $\varphi \in \Phi$ and all θ_{kl} with $k < l$ satisfy the equalities

$$\begin{aligned}\varphi(t_{ij}) &= t_{i\varphi(j)}, & \theta_{kl}(t_{ij}) &= t_{ij} \quad (j \neq l), \\ \theta_{kl}(t_{il}) &= t_{ik}t_{il} + t_{ik} + t_{il}.\end{aligned}\tag{4}$$

Let K be an associative and commutative ring with identity, set $R_c(K) = K[t_{ij} | i = 1, \dots, c; j \in \mathbf{N}]$, and let $\tilde{R}_c(K)$ be the K -subalgebra of $R_c(K)$ generated by all polynomials of the form

$$(1 + k_1 t_1^{(1)} + \dots + k_l t_1^{(l)}) \cdots (1 + k_1 t_c^{(1)} + \dots + k_l t_c^{(l)}),\tag{5}$$

where $l \geq 0$, $k_1, \dots, k_l \in \mathbf{Z}$, and $t_i^{(j)} = t_{i1}^{n_{j1}} t_{i2}^{n_{j2}} \cdots$ are monomials with coefficient unity ($1 \leq i \leq c$, $1 \leq j \leq l$). We stress that the coefficients k_i in (5) can take on integer values only, not arbitrary values in K . Clearly, $\tilde{R}_c(\mathbf{Z}) = \tilde{R}_c$. We define the families Φ and Θ of endomorphisms φ and θ_{kl} ($k < l$) of $R_c(K)$ via the formulas (4).

Let N_1, N_2 , and N_3 be arbitrary fixed integers, with $N_1, N_2 > 0$ and $N_3 \geq 0$. Let $V = V_{N_1, N_2, N_3}$ be the set of all sequences $v = \{(j_{l1}, j_{l2})\}$ of ordered pairs of integers such that $0 \leq j_{l1} \leq N_1 - 1$ and $0 \leq j_{l2} \leq N_2 - 1$ for all $l \in \mathbf{N}$, $(j_{l1}, j_{l2}) = (0, 0)$ for almost all values of l , and not more than N_3 of the j_{l2} ($l \in \mathbf{N}$) are nonzero. Let $M_c = M_{c, N_1, N_2, N_3}(K)$ be the free $R_c(K)$ -module freely generated by the elements v of V_{N_1, N_2, N_3} . For every $\varphi: \mathbf{N} \rightarrow \mathbf{N}$ in Φ , we define a map of V to itself, also denoted by φ , by the rule $\varphi(\{(j_{l1}, j_{l2})\}) = \{(j'_{l1}, j'_{l2})\}$, where $j'_{lm} = j_{km}$ if $l = \varphi(k)$, and $j'_{lm} = 0$ if $l \notin \varphi(\mathbf{N})$ (here $m = 1, 2$). For arbitrary $k, l \in \mathbf{N}$ with $k < l$, we define a map $\theta_{kl}: V \rightarrow V$ by the rule $\theta_{kl}(\{(j_{m1}, j_{m2})\}) = \{(j'_{m1}, j'_{m2})\}$, where $j'_{kl} = j_{l1}$, $j'_{k2} = 0$, and $j'_{mn} = j_{mn}$ for $m \neq l$ and $n = 1, 2$. Let Λ be the set of all automorphisms λ of the ring $R_c(K)$ (of semilinear maps of the module L_c , maps of V to itself) of the form

$$\lambda = \theta_{k_{11}k_{12}} \theta_{k_{21}k_{22}} \cdots \theta_{k_{s1}k_{s2}} \varphi,$$

where $s \geq 0$, $k_{j1} \notin \varphi(\mathbf{N})$ ($j = 1, \dots, c$), and $k_{i1} \neq k_{j1}$ for $i \neq j$.

It is not hard to check that Λ is a semigroup. We use Λ to denote the semigroup of semilinear transformations λ of the module M_c such that $\lambda(v \cdot r) = \lambda(v) \cdot \lambda(r)$ ($v \in V$, $r \in R_c(K)$).

PROPOSITION 1.12. *For all integers c, N_1, N_2 , and N_3 with $c, N_1, N_2 > 0$ and $N_3 \geq 0$, and every commutative associative Noetherian ring K with 1, ascending chains of $\tilde{R}_c(K)$ -submodules of $M_{c, N_1, N_2, N_3}(K)$ admitting all endomorphisms in Λ break off.*

Since it is not hard to define an injective homomorphism χ of the $R_c(\mathbb{Z})$ -module L_c to $M_{c, 1, 2^{2c}, 2c}(\mathbb{Z})$ such that $\chi(\lambda(w)) = \lambda(\chi(w))$ for all $\lambda \in \Lambda$ and all $w \in L_c$, it follows from Proposition 1.12 that ascending chains of $\tilde{R}_c$ -submodules of L_c admitting all the maps from Λ break off. A fortiori, Proposition 1.11 and thus Theorem 1' now follow.

REMARK ON THE PROOF OF THEOREM 2'. Let $L'_{n-k}(K)$ be the free module over $R_n(K)$ freely generated by the elements $w_{i_1 \dots i_{2n-2}}$, for $i_1, \dots, i_{2n-2} \in \mathbb{N}$. Just as in the proof of Theorem 1.3 of [14], it can be shown that Theorem 2' will follow if it can be established that ascending chains of $\tilde{R}_n(K)$ -submodules of $L'_{n-k}(K)$ admitting all the maps from Φ and Θ break off. However, it is not hard to deduce from Proposition 1.12 that such chains of submodules of $L'_{n-k}(K)$ do in fact break off, just as it was for L_c .

§2. The Noetherian property for a module over a polynomial ring: reduction to the case of a ring of finite characteristic

In §1.2 we showed that Theorems 1' and 2' reduce to Proposition 1.12; here we shall establish that proposition under the assumptions that it holds for all Noetherian commutative associative rings K with identity such that $(c!)^{c^2+1}K = 0$.

1. Let $\overline{R}_c(K)$ be the subalgebra of $R_c(K)$ consisting of all polynomials g such that

$$\begin{aligned} g(t_{11}, t_{12}, \dots; \dots; t_{c1}, t_{c2}, \dots) \\ = g(t_{\tau(1)1}, t_{\tau(1)2}, \dots; \dots; t_{\tau(c)1}, t_{\tau(c)2}, \dots) \end{aligned}$$

for every permutation τ of $1, \dots, c$. Clearly, $\tilde{R}_c(K) \subseteq \overline{R}_c(K)$. For brevity, in what follows we shall refer to $\overline{R}_c(K)$ -submodules ($\tilde{R}_c(K)$ -submodules respectively) of $M_{c, N_1, N_2, N_3}(K)$ admitting the transformations in Λ as $(\overline{R}_c(K) \cup \Lambda)$ -submodules (as $(\tilde{R}_c(K) \cup \Lambda)$ -submodules respectively).

The following lemma is proved in the same way as Lemma 3.1 of [14], and has an important part to play in the proof.

LEMMA 2.1. *For arbitrary integers c, N_1, N_2 , and N_3 with $N_3 \geq 0$ and $c, N_1, N_2 > 0$ and every Noetherian commutative associative ring K with 1, ascending chains of $(\overline{R}_c(K) \cup \Lambda)$ -submodules of $M_{c, N_1, N_2, N_3}(K)$ break off.*

If K is a ring such that $(c!)^{-1} \in K$, Proposition 1.12 follows from the lemma since in that case $\tilde{R}_c(K) = \overline{R}_c(K)$: this is proved below in Lemma 2.3.

The following proposition will be proved in §3.

PROPOSITION 2.2. *For arbitrary integers c, N_1, N_2 , and N_3 with $c, N_1, N_2 > 0$ and $N_3 \geq 0$ and every Noetherian commutative associative ring K with 1 such that $(c!)^{c^2+1}K = 0$, ascending chains of $(\tilde{R}_c(K) \cap \Lambda)$ -submodules of $M_{c, N_1, N_2, N_3}(K)$ break off.*

We shall prove Proposition 1.12 under the assumption that Proposition 2.2 holds.

LEMMA 2.3. Set $I_c = (c!)^{c^2+1} \overline{R}_c(K)$. Then:

- 1) $I_c \subseteq \tilde{R}_c(K)$.
- 2) I_c is an ideal of $\overline{R}_c(K)$ closed under all maps from Λ .

PROOF. Item 2) is clear. Let us prove 1). Let S_c be the symmetric group on $\{1, \dots, c\}$. Set

$$t_j^{(i)} = t_{j1}^{n_{i1}} t_{j2}^{n_{i2}} \dots \quad (1 \leq i, j \leq c), \quad (6)$$

where $n_{il} \geq 0$ for $1 \leq i \leq c$ and $l \in \mathbf{N}$. Let

$$\langle t_1^{(1)} t_2^{(2)} \dots t_c^{(c)} \rangle \quad (7)$$

be the sum of all monomials obtained from $t_1^{(1)} \dots t_c^{(c)}$ by permuting the lower indices; that is,

$$\langle t_1^{(1)} \dots t_c^{(c)} \rangle = \sum_{\tau \in S_c : \text{St}} t_{\tau(1)}^{(1)} \dots t_{\tau(c)}^{(c)},$$

where the sum extends over the permutations τ taken one each from the left cosets $\tau \cdot \text{St}$ of the stabilizer St of $t_1^{(1)} \dots t_c^{(c)}$ in S_c . It is clear that

$$\sum_{\tau \in S_c} t_{\tau(1)}^{(1)} \dots t_{\tau(c)}^{(c)} = |\text{St}| \langle t_1^{(1)} \dots t_c^{(c)} \rangle. \quad (8)$$

Let Δ be the Vandermonde determinant of order $c+1$, $\Delta = \det \|a_{ij}\|$, where $a_{ij} = i^j$ for $0 \leq i, j \leq c$ (we set $0^0 = 1$ in this situation). Let $t_1^{(j)}$ ($1 \leq i, j \leq c$) be any polynomials of the form (6) such that $t_i^{(j)} \neq 1$ for $1 \leq j \leq k$ and $t_i^{(j)} = 1$ for $k+1 \leq j \leq c$. Set

$$f_{\alpha_1, \dots, \alpha_k} = \prod_{i=1}^c (1 + \alpha_1 t_1^{(1)} + \dots + \alpha_k t_i^{(k)}) \quad (0 \leq \alpha_1, \dots, \alpha_k \leq c),$$

so that $f_{\alpha_1, \dots, \alpha_k} \in \tilde{R}_c(K)$ for all $\alpha_1, \dots, \alpha_k$, and let f be the sum of all the monomials occurring in $f_{\alpha_1, \dots, \alpha_k}$ with coefficient $\alpha_1 \dots \alpha_k$. Clearly

$$f = \sum_{\tau \in S_c} t_{\tau(1)}^{(1)} \dots t_{\tau(k)}^{(k)};$$

and it is not difficult to show that the polynomial $\Delta^k \cdot f$ can be written as a $\mathbf{Z}$ -linear combination of polynomials $f_{\alpha_1, \dots, \alpha_k}$ ($0 \leq \alpha_1, \dots, \alpha_k \leq c$), so that $\Delta^k \cdot f \in \tilde{R}_c(K)$. Using (8), we get that

$$|\text{St}| \cdot \Delta^k \langle t_1^{(1)} \dots t_k^{(k)} \rangle \in \tilde{R}_c(K),$$

where St is the stabilizer of $t_1^{(1)} \dots t_k^{(k)}$ in S_c ; since Δ and $|\text{St}|$ divide $(c!)^c$ and $c!$ respectively, and $k \leq c$, we have

$$(c!)^{c^2+1} \langle t_1^{(1)} \dots t_k^{(k)} \rangle \in \tilde{R}_c(K).$$

Finally, since $1 \in \tilde{R}_c(K)$ and all polynomials of the form (7)—including the identity—form a K -basis of the algebra $\overline{R}_c(K)$, it follows that $(c!)^{c^2+1} \overline{R}_c \subseteq \tilde{R}_c$. This completes the proof of Lemma 2.3.

LEMMA 2.4. If $M^{(1)} \subseteq M^{(2)} \subseteq \dots$ is an infinite ascending chain of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules in $M_c = M_{c, N_1, N_2, N_3}(K)$, then there exist an $(\overline{R}_c(K) \cup \Lambda)$ -submodule $\widehat{M}$ of M_c and a natural number k such that

- 1) $M^{(i)} \subseteq \widehat{M}$ ($i = 1, 2, \dots$), and
- 2) $\widehat{M} \cdot I_c \subseteq M^{(i)}$ ($i = k, k+1, \dots$).

PROOF. Let $M^{(1)} \subseteq M^{(2)} \subseteq \dots$ be an ascending chain of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of M_c ; then $M^{(1)} \cdot I_c \subseteq M^{(2)} \cdot I_c \subseteq \dots$ is an ascending chain of $(\overline{R}_c(K) \cup \Lambda)$ -submodules of M_c , and by Lemma 2.1 there exists a natural number k such that $M^{(k)} \cdot I_c = M^{(k+1)} \cdot I_c = \dots$.

Set $M = M^{(k)} \cdot I_c$; we write

$$\widehat{M} = \{m \in M_c \mid (c!)^{c^2+1} \cdot m \in M\}.$$

It is clear that $\widehat{M}$ is an $(\overline{R}_c(K) \cup \Lambda)$ -submodule of M_c . Since for every natural number i we have $(c!)^{c^2+1} M^{(i)} \subseteq M^{(i)} \cdot I_c \subseteq M$, it follows that $M^{(i)} \subseteq \widehat{M}$ for $i \in \mathbb{N}$, and 1) holds. On the other hand, $\widehat{M} \cdot I_c = (c!)^{c^2+1} \widehat{M} \subseteq M$, and since $I_c \subseteq \tilde{R}_c(K)$ by Lemma 2.3, for $i \geq k$ we have $M = M^{(i)} \cdot I_c \subseteq M^{(i)} \tilde{R}_c(K) = M^{(i)}$. This means that $\widehat{M} \cdot I_c \subseteq M \subseteq M^{(i)}$ for $i = k, k+1, \dots$; that is, 2) holds and we have completed the proof of the lemma.

LEMMA 2.5. Let c, N_1, N_2, N_3 , and N_4 be integers such that $c, N_1, N_2, N_4 > 0$ and $N_3 \geq 0$. Take N_4 copies of the set V_{N_1, N_2, N_3} , denoted by $V_{N_1, N_2, N_3}^1, V_{N_1, N_2, N_3}^2, \dots, V_{N_1, N_2, N_3}^{N_4}$. Let K be a Noetherian commutative associative ring with 1, and let $\overline{M}_{c, N_4} = \overline{M}_{c, N_1, N_2, N_3, N_4}(K)$ be the free module over $\overline{R}_c(K)$ freely generated by the elements $v^{(i)} \in V_{N_1, N_2, N_3}^i$, $1 \leq i \leq N_4$. Then ascending chains of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of $\overline{M}_{c, N_4} / \overline{M}_{c, N_4} \cdot I_c$ break off.

PROOF. Let $\overline{M}_c$ be the free $\overline{R}_c(K)$ -module freely generated by the elements $v \in V_{N_1, N_2, N_3}$; then $\overline{M}_{c, N_4}$ is a direct sum of finitely many $\overline{R}_c(K)$ -modules, each of which is isomorphic (as an $(\overline{R}_c(K) \cup \Lambda)$ -module) to $\overline{M}_c$. Thus, it is enough to show that ascending chains of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of $\overline{M}_c / \overline{M}_c \cdot I_c$ break off. Set $M_c = M_{c, N_1, N_2, N_3}(K)$; clearly, $\overline{M}_c / \overline{M}_c \cdot I_c$ is embedded (as an $(\tilde{R}_c(K) \cup \Lambda)$ -module) in $M_c / M_c \cdot I_c$. All that remains now is to note that ascending chains of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of $M_c / M_c \cdot I_c$ break off (Proposition 2.2). This completes the proof of Lemma 2.5.

2. We embark at once on the proof that ascending chains of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of $M_c = M_{c, N_1, N_2, N_3}(K)$ break off. Let $M^{(1)} \subseteq M^{(2)} \subseteq \dots$ be any infinite ascending chain of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of M_c . Let $\widehat{M}$ be the $(\overline{R}_c(K) \cup \Lambda)$ -submodule of M_c whose existence is guaranteed by Lemma 2.4. By Lemma 2.1, $\widehat{M}$ is a finitely generated $(\overline{R}_c(K) \cup \Lambda)$ -module.

Let $m^{(1)}, \dots, m^{(N_4)}$ be $(\overline{R}_c(K) \cup \Lambda)$ -module generators for $\widehat{M}$; then, as $\overline{R}_c(K)$ -module, $\widehat{M}$ is generated by the elements of the form $\lambda(m^{(i)})$ with $1 \leq i \leq N_4$ and

$\lambda \in \Lambda$. Set $m^{(i)} = \sum_l v_{il} r_{il}$, where $r_{il} \in R_c(K)$ and $v_{il} \in V_{N_1, N_2, N_3}$ for all i and l . Let N' be a natural number such that (i) for all i and l , r_{il} fails to contain the unknowns $t_{1j}, \dots, t_{cj}$ for $j > N'$, and (ii) all terms in the sequence v_{il} with indices greater than N' are $(0, 0)$.

Set

$$\bar{v} = ((1, 1), \dots, (N', N'), (0, 0), \dots) \in V_{N'+1, N'+1, N'+1}$$

and let $V_{N'+1, N'+1, N'+1}^i$, $1 \leq i \leq N_4$, be N_4 copies of $V_{N'+1, N'+1, N'+1}$. Denote by $\bar{v}^{(i)}$, $1 \leq i \leq N_4$, the element $\bar{v}$ taken in the i th copy, that is, in $V_{N'+1, N'+1, N'+1}^i$. Consider the $(\bar{R}_c(K) \cup \Lambda)$ -submodule $\bar{M}$ of $\bar{M}_{c, N_4} = \bar{M}_{c, N'+1, N'+1, N'+1, N_4}(K)$ generated by the elements $\bar{v}^{(i)}$, $1 \leq i \leq N_4$; as an $\bar{R}_c(K)$ -module, $\bar{M}$ is generated by the elements $\lambda(\bar{v}^{(i)})$ for $\lambda \in \Lambda$ and $1 \leq i \leq N_4$.

Note that if an element r of $R_c(K)$ contains no occurrence of the unknowns t_{ij} with $j > N'$, and in a sequence v in V_{N_1, N_2, N_3} all terms with indices greater than N' are $(0, 0)$, then for arbitrary $\lambda_1, \lambda_2 \in \Lambda$ it follows that $\lambda_1(r) = \lambda_2(r)$ and $\lambda_1(v) = \lambda_2(v)$ whenever $\lambda_1(\bar{v}) = \lambda_2(\bar{v})$. Thus, if $\lambda_1, \lambda_2 \in \Lambda$, $1 \leq i \leq N_4$, and $\lambda_1(\bar{v}^{(i)}) = \lambda_2(\bar{v}^{(i)})$, we have $\lambda_1(m^{(i)}) = \lambda_2(m^{(i)})$; and then we can make a good definition of a homomorphism $\pi: \bar{M} \rightarrow \widehat{M}$ of $\bar{R}_c(K)$ -modules by setting $\pi(\lambda(\bar{v}^{(i)})) = \lambda(m^{(i)})$ for $1 \leq i \leq N_4$ and every $\lambda \in \Lambda$. It is clear that the inverse image in $\bar{M}$ of an $(\tilde{R}_c(K) \cup \Lambda)$ -submodule of $\widehat{M}$ is an $(\tilde{R}_c(K) \cup \Lambda)$ -submodule of $\bar{M}$. In particular, $\pi^{-1}(M^{(1)}) \subseteq \pi^{-1}(M^{(2)}) \subseteq \dots$ is an infinite ascending chain of $(\tilde{R}_c(K) \cap \Lambda)$ -submodules of $\bar{M}$. Further, it is clear that $\bar{M} \cdot I_c \subseteq \pi^{-1}(\widehat{M} \cdot I_c)$; since $\widehat{M} \cdot I_c \subseteq M^{(i)}$ for $i = k, k+1, \dots$, it follows that $\pi^{-1}(\widehat{M} \cdot I_c) \subseteq \pi^{-1}(M^{(i)})$, which implies that $\bar{M} \cdot I_c \subseteq \pi^{-1}(M^{(i)})$ for $i = k, k+1, \dots$. Since $\bar{M}/\bar{M} \cdot I_c$ is isomorphically embedded in $\bar{M}_{c, N_4}/\bar{M}_{c, N_4} \cdot I_c$, Lemma 2.5 gives that the chain $\pi^{-1}(M^{(k)}) \subseteq \pi^{-1}(M^{(k+1)}) \subseteq \dots$ breaks off. This means that the chain $M^{(1)} \subseteq M^{(2)} \subseteq \dots$ also breaks off.

Therefore, we have established Proposition 1.12 under the assumption that Proposition 2.2 holds.

§3. The Noetherian property for a module over a polynomial ring: the case where the ring is of finite characteristic

In the preceding section, the proofs of Theorems 1' and 2' were reduced to proving Proposition 2.1; we shall give that proof here. Proposition 2.2 is a special case of Proposition 1.12, obtained from the latter under the additional assumption that $(c!)^{c^2+1}K = 0$.

Thus, suppose that K is a Noetherian commutative associative ring with 1 such that $(c!)^{c^2+1}K = 0$. Let $(c!)^{c^2+1} = p_1^{\alpha_1} \dots p_s^{\alpha_s}$ be the decomposition of $(c!)^{c^2+1}$ into its primary components; it is clear that $p_i \leq c$ for $i = 1, \dots, s$. Note that ascending chains of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of $M_{c, N_1, N_2, N_3}(K)$ break off if the same is true for the modules $M_{c, N_1, N_2, N_3}(K/p_i^{\alpha_i}K)$ ($i = 1, \dots, s$); and this, in turn, is true if chains in $M_{c, N_1, N_2, N_3}(K/p_iK)$ ($i = 1, \dots, s$) break off. Thus, everywhere in what follows we shall assume that $pK = 0$ for some prime $p \leq c$.

The proof that ascending chains of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of $M_{c, N_1, N_2, N_3}(K)$ break off when K is a ring of prime characteristic $p \leq c$ will proceed by induction on c . Assume that ascending chains of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of $M_{c', N'_1, N'_2, N'_3}(K)$ break off for all c' with $p \leq c' < c$ and all N'_1, N'_2 , and N'_3 (for $c' < p$ such chains break off by Lemma 2.1); we claim that chains in $M_{c, N_1, N_2, N_3}(K)$ break off.

LEMMA 3.1. *Let K be a commutative associative ring with 1 of prime characteristic $p \leq c$. Let J_c be the linear span in $R_c(K)$ of all polynomials $\langle t_1^{(1)} \dots t_c^{(c)} \rangle$ (see the proof of Lemma 2.3) such that the following conditions both hold:*

- 1) $t_j^{(i)} = t_{j_1}^{n_{i1}} t_{j_2}^{n_{i2}} \dots$ ($1 \leq i, j \leq c$), where $t_j^{(i)} \neq 1$ for all i .
- 2) p does not divide the order of the stabilizer of the monomial $t_1^{(1)} \dots t_c^{(c)}$ in the symmetric group S_c on $\{1, \dots, c\}$ (that is, there are not p coincidences among the monomials $t_i^{(i)}$, $i = 1, \dots, c$).

Then the following assertions are true:

- 1) $J_c \subseteq \tilde{R}_c(K)$.
- 2) J_c is an ideal in $\bar{R}_c(K)$ admitting all the mappings from Λ .

PROOF. We first prove that $J_c \subseteq \tilde{R}_c(K)$. Let $t_j^{(i)} = t_{j_1}^{n_{i1}} t_{j_2}^{n_{i2}} \dots$ ($1 \leq i, j \leq c$) be any monomials such that $t_j^{(i)} \neq 1$ for all i . Set

$$g = \sum_{\tau \in S_c} t_{\tau(1)}^{(1)} t_{\tau(2)}^{(2)} \dots t_{\tau(c)}^{(c)}$$

and

$$f_{i_1, \dots, i_k} = \prod_{j=1}^c (1 + t_j^{(i_1)} + \dots + t_j^{(i_k)}) \quad (1 \leq i_1 < \dots < i_k \leq c, \quad 0 \leq k \leq c).$$

The following identity is readily checked:

$$g = f_{1, 2, \dots, c} - \dots + (-1)^q \sum_{1 \leq i_1 < \dots < i_{c-q} \leq c} f_{i_1, \dots, i_{c-q}} + \dots + (-1)^{c-1} \sum_{1 \leq i \leq c} f_i + (-1)^c.$$

Since $f_{i_1, \dots, i_k} \in \tilde{R}_c(K)$ for all k and $i_1, \dots, i_k$, it follows that $g \in \tilde{R}_c(K)$. Note that $g = |\text{St}| \langle t_1^{(1)} \dots t_c^{(c)} \rangle$, where St is the stabilizer of $t_1^{(1)} \dots t_c^{(c)}$ in S_c . If the characteristic p of K does not divide the order of St , then $|\text{St}|$ is invertible in K and $\langle t_1^{(1)} \dots t_c^{(c)} \rangle \in \tilde{R}_c(K)$. Thus, $J_c \subseteq \tilde{R}_c(K)$.

We claim next that J_c is an ideal of $\bar{R}_c(K)$ admitting all endomorphisms in Λ . Let $\chi_1: R_c(K) \rightarrow R_{c-1}(K)$ be an algebra homomorphism such that $\chi_1(t_{ij}) = t_{ij}$ for $i < c$ and all j , and $\chi_1(t_{cj}) = 0$ for all j . Clearly, $\chi_1(\lambda(r)) = \lambda(\chi_1(r))$ for all $\lambda \in \Lambda$ and $r \in R_c(K)$. Let $\bar{\chi}_1$ be the restriction of χ_1 to $\bar{R}_c(K)$, and $J_{c,1}$ the kernel of $\bar{\chi}_1$. It is clear that the set of all polynomials $\langle t_1^{(1)} \dots t_c^{(c)} \rangle$ such that $t_j^{(i)} \neq 1$ for all i and j is a K -basis for $J_{c,1}$. It is also clear that $J_{c,1}$ is an ideal of $\bar{R}_c(K)$ admitting all the endomorphisms of Λ .

Let $\chi_2: R_c(K) \rightarrow R_{c-p+1}(K)$ be an algebra homomorphism such that $\chi_2(t_{ij}) = t_{ij}$ for $i \leq c-p$ and all j , and $\chi_2(t_{ij}) = t_{(c-p+1)j}$ for all $i \geq c-p+1$ and all j . Clearly, $\chi_2(\lambda(r)) = \lambda(\chi_2(r))$ for all $\lambda \in \Lambda$ and $r \in R_c(K)$. Let $\bar{\chi}_2$ be the restriction of χ_2 to $\bar{R}_c(K)$, and $J_{c,2}$ the kernel of $\bar{\chi}_2$. It is clear that $J_{c,2}$ is an ideal of $\bar{R}_c(K)$

admitting all endomorphisms from Λ . It is not difficult to show that a K -basis for $J_{c,2}$ can be chosen so as to consist of the polynomials of the form $\langle t_1^{(1)} \cdots t_c^{(c)} \rangle$ with $t_j^{(i)} = t_{j1}^{n_{i1}} t_{j2}^{n_{i2}} \cdots$ ($1 \leq i, j \leq c$), and perhaps some of the $t_j^{(i)}$ are 1, but no p of the monomials $t_1^{(1)}, \dots, t_c^{(c)}$ are equal (equivalently, the order of the stabilizer of $t_1^{(1)} \cdots t_c^{(c)}$ in S_c is not divisible by p).

Now all we do is to observe that the set of all polynomials of the form $\langle t_1^{(1)} \cdots t_c^{(c)} \rangle$, where $t_j^{(i)} = t_{j1}^{n_{i1}} t_{j2}^{n_{i2}} \cdots$ ($1 \leq i, j \leq c$), such that $t_j^{(i)} \neq 1$ for all i and j and no p of the polynomials $t_1^{(1)}, \dots, t_c^{(c)}$ are equal, is a K -basis for $J_{c,1} \cap J_{c,2}$. This means that $J_{c,1} \cap J_{c,2} = J_c$, so that J_c is an ideal of $\bar{R}_c(K)$ admitting all maps from Λ . This completes the proof of Lemma 3.1.

LEMMA 3.2. *Let K be a Noetherian commutative associative ring with 1 of prime characteristic $p \leq c$, and $M^{(1)} \subseteq M^{(2)} \subseteq \cdots$ an infinite ascending chain of $(\bar{R}_c(K) \cup \Lambda)$ -submodules of $M_c = M_{c, N_1, N_2, N_3}(K)$. Then there exist an $(\bar{R}_c(K) \cup \Lambda)$ -submodule $\widehat{M}$ of M_c and a natural number k such that*

- 1) $M^{(i)} \subseteq \widehat{M}$ ($i \in \mathbb{N}$), and
- 2) $\widehat{M} \cdot J_c \subseteq M^{(i)}$ ($i = k, k+1, \dots$).

The proof of this lemma goes just like that of Lemma 2.4, if in the latter we replace I_c by J_c and define $\widehat{M}$ as follows:

$$\widehat{M} = \{m \in M_c \mid (\forall \lambda \in \Lambda) (\forall r \in J_c) (\lambda(m) \cdot r \in M)\}.$$

LEMMA 3.3. *Let K be a Noetherian commutative associative ring with 1 of characteristic $p \leq c$. Let N_1, N_2, N_3 , and N_4 be integers with $N_1, N_2, N_4 > 0$ and $N_3 \geq 0$. Take N_4 copies of V_{N_1, N_2, N_3} , to be denoted by $V_{N_1, N_2, N_3}^1, \dots, V_{N_1, N_2, N_3}^{N_4}$. Set $\overline{M}_{c, N_4} = \overline{M}_{c, N_1, N_2, N_3, N_4}(K)$ for the free module over $\bar{R}_c(K)$ freely generated by the elements $v^{(i)} \in V_{N_1, N_2, N_3}^i$ ($1 \leq i \leq N_4$). Then ascending chains of $(\bar{R}_c(K) \cup \Lambda)$ -submodules of $\overline{M}_{c, N_4} / \overline{M}_{c, N_4} \cdot J_c$ break off.*

PROOF. Let $\overline{M}_c$ be the free $\bar{R}_c(K)$ -module freely generated by the elements v of V_{N_1, N_2, N_3} , so that $\overline{M}_c \subseteq M_c$. Clearly, to prove the lemma it is enough to establish that ascending chains of $(\bar{R}_c(K) \cup \Lambda)$ -submodules of $\overline{M}_c / \overline{M}_c \cdot J_c$ break off.

Set $M_d = M_{d, N_1, N_2, N_3}(K)$ ($d \in \mathbb{N}$), and let $\chi_1: M_c \rightarrow M_{c-1}$ be a semilinear map such that $\chi_1(v \cdot r) = v \cdot \chi_1(r)$ for all $v \in V_{N_1, N_2, N_3}$ and $r \in R_c(K)$ (see the proof of Lemma 3.1 for the definition of the homomorphism $\chi_1: R_c(K) \rightarrow R_{c-1}(K)$). Clearly, $\chi_1(\lambda(m)) = \lambda(\chi_1(m))$ for all $\lambda \in \Lambda$ and $m \in M_c$. Note also that $\chi_1(\bar{R}_c(K)) = \bar{R}_{c-1}(K)$; therefore, for every $(\bar{R}_c(K) \cup \Lambda)$ -submodule M of M_c , $\chi_1(M)$ is an $(\bar{R}_{c-1}(K) \cup \Lambda)$ -submodule of M_{c-1} .

Let $\bar{\chi}_1$ be the restrictions of χ_1 to $\overline{M}_c$; then $\text{Ker } \bar{\chi}_1 = \overline{M}_c \cdot J_{c,1}$; since ascending chains of $(\bar{R}_{c-1}(K) \cup \Lambda)$ -submodules of M_{c-1} break off (by induction if $c-1 \geq p$, and by Lemma 2.1 if $c-1 < p$), ascending chains of $(\bar{R}_c(K) \cup \Lambda)$ -submodules of $\overline{M}_c / \overline{M}_c \cdot J_{c,1}$ break off.

Let $\chi_2: M_c \rightarrow M_{c-p+1}$ be a semilinear map such that $\chi_2(v \cdot r) = v \cdot \chi_2(r)$ for all

$v \in V_{N_1, N_2, N_3}$ and $r \in R_c(K)$ (see the proof of Lemma 3.1 for the definition of the homomorphism $\chi_2: R_c(K) \rightarrow R_{c-p+1}(K)$). Clearly, $\chi_2(\lambda(m)) = \lambda(\chi_2(m))$ for all $\lambda \in \Lambda$ and $m \in M_c$.

Set $R'_{c-p+1}(K) = K[t_{ij}, t_{(c-p+1)j}^p | 1 \leq i \leq c-p, j \in \mathbb{N}]$, so that $R'_{c-p+1}(K) \subseteq R_{c-p+1}(K)$; then, as is not hard to check, $\chi_2(\bar{R}_c(K)) \subseteq R'_{c-p+1}(K)$. Let $\eta: R_{c-p+1}(K) \rightarrow R'_{c-p+1}(K)$ be an endomorphism of $R_{c-p+1}(K)$ such that $\eta(t_{ij}) = t_{ij}$ for $1 \leq i \leq c-p$ and $j \in \mathbb{N}$, and $\eta(t_{(c-p+1)j}^p) = t_{(c-p+1)j}^p$ for $j \in \mathbb{N}$. Since $\eta(\theta_{kl}(t_{ij})) = \theta_{kl}(\eta(t_{ij}))$ for all i, j, k , and l with $k < l$, we have $\eta(\lambda(r)) = \lambda(\eta(r))$ for all $\lambda \in \Lambda$ and $r \in R_{c-p+1}(K)$.

Let $\tilde{R}'_{c-p+1}(K)$ be the subalgebra of $R'_{c-p+1}(K)$ generated by the polynomials

$$(1 + k_1 t_1^{(1)} + \dots + k_l t_1^{(l)}) \dots (1 + k_1 t_{c-p}^{(1)} + \dots + k_l t_{c-p}^{(l)})(1 + k_1 (t_{c-p+1}^{(1)})^p + \dots + k_l (t_{c-p+1}^{(l)})^p),$$

where $l \geq 0$, $k_1, \dots, k_l \in \mathbb{Z}$, and $t_j^{(i)} = t_{j1}^{n_1} t_{j2}^{n_2} \dots$ are nonidentity monomials for $1 \leq i \leq l$ and $1 \leq j \leq c-p+1$. Clearly, $\chi_2(\tilde{R}_c(K)) = \eta(\tilde{R}_{c-p+1}(K)) = \tilde{R}'_{c-p+1}(K)$.

Let M'_{c-p+1} be the free module over $R'_{c-p+1}(K)$ freely generated by the elements v of V_{N_1, N_2, N_3} ; then $\chi_2(\bar{M}_c) \subseteq M'_{c-p+1}$. Let η be a semilinear map from M_{c-p+1} to M'_{c-p+1} such that $\eta(v \cdot r) = v \cdot \eta(r)$ for all $v \in V_{N_1, N_2, N_3}$ and all $r \in R_{c-p+1}(K)$. Then $\eta(\lambda(m)) = \lambda(\eta(m))$ for all $\lambda \in \Lambda$ and $m \in M_{c-p+1}$. By the above remarks, if M is an $(\tilde{R}'_{c-p+1}(K) \cup \Lambda)$ -submodule of M'_{c-p+1} then $\eta^{-1}(M)$ is an $(\tilde{R}_{c-p+1}(K) \cup \Lambda)$ -submodule of M_{c-p+1} ; since ascending chains of $(\tilde{R}_{c-p+1}(K) \cup \Lambda)$ -submodules of M_{c-p+1} break off (by induction if $c-p+1 \geq p$, and by Lemma 2.1 if $c-p+1 < p$), it follows that ascending chains of $(\tilde{R}'_{c-p+1}(K) \cup \Lambda)$ -submodules of M'_{c-p+1} break off.

Let $\bar{\chi}_2$ be the restriction of χ_2 to $\bar{M}_c$; then $\text{Ker } \bar{\chi}_2 = \bar{M}_c \cdot J_{c,2}$. By the above remarks, if M is an $(\tilde{R}_c(K) \cup \Lambda)$ -submodule of $\bar{M}_c$, then $\bar{\chi}_2(M)$ is an $(\tilde{R}'_{c-p+1}(K) \cup \Lambda)$ -submodule of M'_{c-p+1} ; since M'_{c-p+1} is a Noetherian $(\tilde{R}'_{c-p+1}(K) \cup \Lambda)$ -module, ascending chains of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of $\bar{M}_c / \bar{M}_c \cdot J_{c,2}$ break off.

Since $J_c = J_{c,1} \cap J_{c,2}$ (see the proof of Lemma 3.1), and $\bar{M}_c$ is a free $\bar{R}_c(K)$ -module, we see that $\bar{M}_c \cdot J_c = \bar{M}_c \cdot J_{c,1} \cap \bar{M}_c \cdot J_{c,2}$; thus $\bar{M}_c / \bar{M}_c \cdot J_c$ is an $(\bar{R}_c(K) \cup \Lambda)$ -submodule of $\bar{M} = \bar{M}_c / \bar{M}_c \cdot J_{c,1} \oplus \bar{M}_c / \bar{M}_c \cdot J_{c,2}$; since $\bar{M}$ is a Noetherian $(\tilde{R}_c(K) \cup \Lambda)$ -submodule by what was said above, it follows that ascending chains of $(\tilde{R}_c(K) \cup \Lambda)$ -submodules of $\bar{M}_c / \bar{M}_c \cdot J_c$ break off. This completes the proof of Lemma 3.3.

The proof of Proposition 2.2 now proceeds by repeating the arguments in §2.2, with J_c replacing I_c , and Lemmas 3.1, 3.2, and 3.3 replacing Lemmas 2.3, 2.4, and 2.5 respectively.

This completes the proof of Propositions 2.2 and 1.12, and therefore of Theorems 1' and 2'.

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